

Pollution and Climate Change

ENGR49580D Technology, Society and Sustainability

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Fossil Fuel Combustion Byproducts

Typical Combustion Byproducts:

- Nitric Oxides (NO_x) – Human Health
- Sulphur (SO_2) – Acid Rain
- Mercury – Neurotoxin
- Ash – Non combustible metals
- Fly Ash – Un-combusted particulate in the smoke stream
- Hydro Carbons – Un-combusted and partially combusted fuel
- Carbon Monoxide – Product of incomplete combustion

NO_x

- Nitric Oxides (NO_x) is the name used for a group of oxides of nitrogen (usually NO and NO₂)
- These are the result of combustion of fuels that contain trace amounts of nitrogen or nitrogen containing compounds
- Reacts with ammonia form nitric acids
- Reacts with VOCs to form ground level ozone, a human health hazard
- NO_x are typically reduced by selective or non-selective catalytic reduction

SO₂

- Sulphur Dioxide (SO₂) is the result of combustion of fuels that contain trace amounts of sulphur or sulphur containing compounds
 - Sulphur compounds may cause smoke to take on a yellow colour
- When dissolved in water, SO₂ forms sulphuric acid, the primary cause of anthropomorphic acid rain
- In liquid fuels (crude oil products) sulphur can be reduced in the production process
- For coal applications, SO₂ must be removed post combustion using technologies know as scrubbers

Mercury

- Three toxic forms
 - Elemental mercury
 - Liquid at room temperature
 - Low boiling point (357°C)
 - Inorganic mercury
 - Organic mercury
 - Typically found in coal fumes
 - Methylmercury (an organic form) accumulates in aquatic ecosystems
 - It is a bio accumulator remaining in higher order predators especially fish

Fly Ash

- Un-combusted solids in the exhaust stream
- Typically present with high carbon content fuels (coal, fuel oil, diesel)
- Results in black smoke
- Typically removed by filtration

Radio-isotopes

Radioisotopes

- Radioisotopes may be present in nature (uranium decay products) or the product of anthropomorphic nuclear reactions
- Radioisotopes may be released into the environment by:
 - Natural processes
 - Improper disposal of radioisotopes used for medical or industrial purposes (including ionizing type smoke detectors)
 - Discharges at nuclear reactors / waste storage facilities
 - Atmospheric detonation of nuclear weapons

Radioisotopes

Isotope lives

- Radioisotope decay rate is measured as a half-life
- Radioisotopes with short half-lives experience radioactive decay quickly and emit many particles / high levels of radiation
- Radioisotopes with long half-lives experience radioactive decay slowly and emit few particles / low levels of radiation
 - These are typically less harmful

Radioisotopes

Decay Products

- A radioactive decay produces 1 or more smaller atoms and energy as a particle or wave (colloquially referred to as radiation)
 - Alpha Particles
 - Highly ionizing (10 to 1000x the chromosomal damage of Beta particles or Gamma rays)
 - Low Penetration – blocked by the outer layers of skin
 - Only a health hazard if the source is ingested or inhaled

Radioisotopes

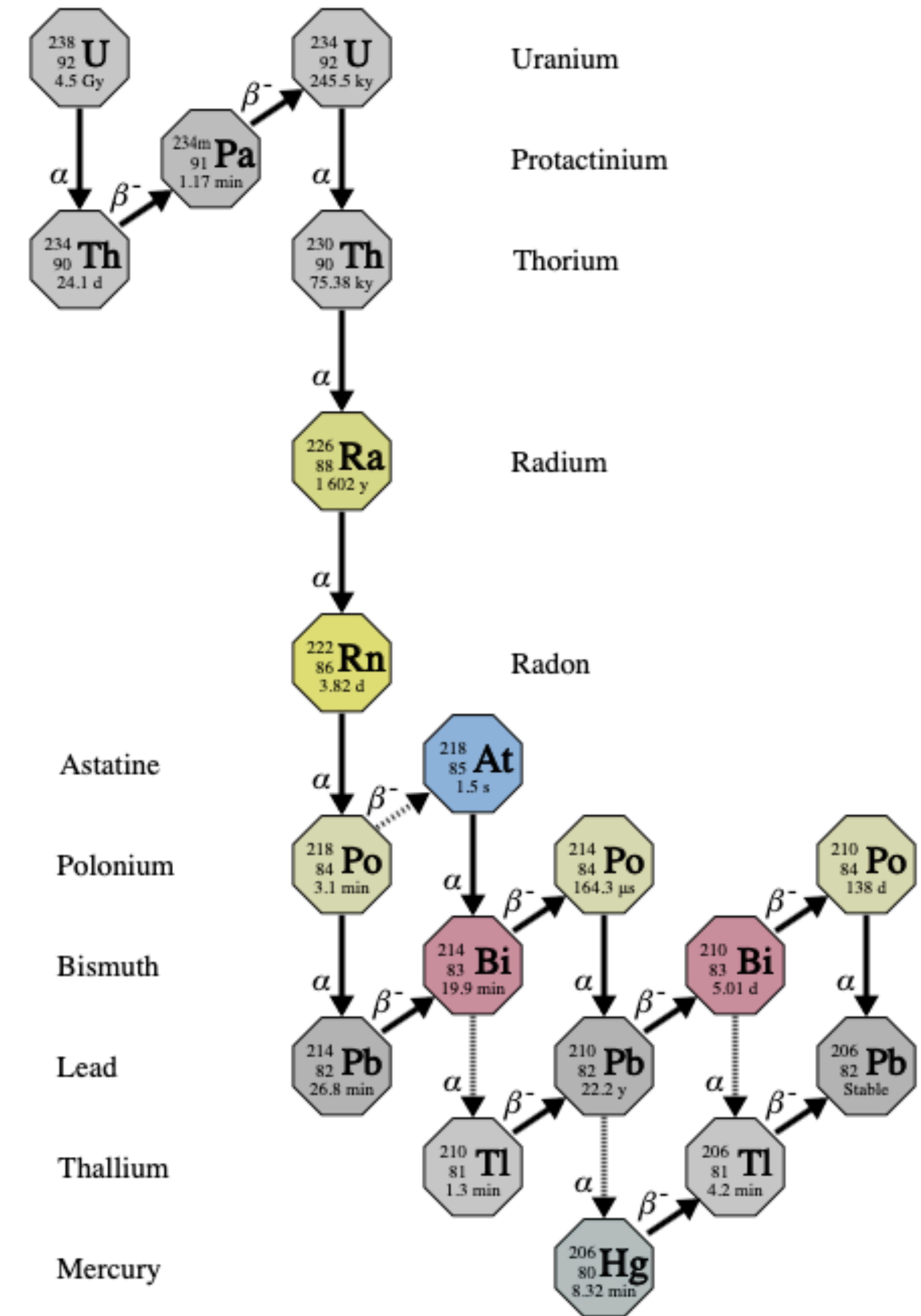
Decay Products

- Beta Particles
 - Medium penetration – blocked by aluminium plate
- Gamma Rays
 - High penetration – requires lead or concrete shielding
- Radiation Injury
 - low does exposure causes increased cancer risk
 - high does exposure causes radiation poisoning symptoms

Radioisotopes

Radium

- Product of the natural decay of Uranium
- Can be extracted from uranium containing ores
- Popular in Industrial and 'Heath' applications in the early 20th century
- Many workplace and intentional exposures
 - Effects of radiation exposure were not understood



Radioisotopes

Radon

- Product of the natural decay of uranium (via thorium then radium)
- Is a noble gas, meaning it does not form atomic bonds with other elements and exists in a gaseous form
- Percolates out of rock formations / soil
- Can also become suspended in ground water
- Absorbed by inhalation
- Leading cause of lung cancer amongst non smokers

Radioisotopes

Strontium-90

- Strontium is a chemical substitute for calcium in many organic processes
 - It is integrated into bones
- ^{90}Sr is a beta emitter with a half life of 28.8 years
- ^{90}Sr integration into bones can increase the chances of leukaemia and bone cancers

Radioisotopes

Iodine-131 and Iodine-129

- ^{131}I is a short-lived fission product with a half-life of 8 days that experiences beta decay
- ^{129}I is a long-lived fission product with a half life of 15 700 years that experiences beta or gamma decay
- Iodine accumulates in the thyroid
- Because of the small size of the thyroid, both forms of radioactive iodine are considered cancer causing
- ^{131}I is used to treat thyroid cancers in high doses

Persistent Organic Pollutants

Persistent Organic Pollutants (POPs)

- This is a regulatory class of pollutants that include:
 - Certain pesticides / insecticides / fungicides
 - PCBs
 - Dioxins and Polychlorinated Dibenzofurans
 - Tetrabromodiphenyl Ether (a precursor to fluoropolymers)
 - Hexabromocyclododecane (HBCD a brominated flame retardant)

Persistent Organic Pollutants (POPs)

Hazards

- These pollutants are long lived in the environment and break down extremely slowly (or not at all) by natural processes
- Many of these pollutants are lipid soluble and tend to bio accumulate
- POPs are associated with:
 - Endocrine disruption
 - Reproductive harm
 - Gestational problems

Persistent Organic Pollutants (POPs)

Regulation

- The Stockholm Convention on Persistent Organic Pollutants seeks to regulate or eliminate approximately 30 POPs
- Under the convention a POP may be:
 - Banned
 - Restricted to specific applications
 - Steps put in place to restrict unintentional production

Halogenated Hydrocarbons

Halogenated Hydrocarbons

- Halogenated Hydrocarbons consist of:
 - Chlorofluorocarbons (CFCs)
 - Hydrochlorofluorocarbons (HCFCs)
 - Hydrofluorocarbons (HFCs)
 - Other fluorine containing gasses may be included such as SF₆
- These compounds are particularly non reactive and therefore persist in the environment

Halogenated Hydrocarbons

Uses

- Refrigerant gasses
- Blowing agents
- Fire suppression gasses
- Electrical Insulators (SF_6)
- Intermediates in the production of other chemicals (eg. Teflon)

Halogenated Hydrocarbons

Hazards

- CFCs
 - Persist in the environment and travel to the upper atmosphere
 - Involved in chemical reactions that deplete ozone levels
- HCFCs are less persistent than CFCs
- HFCs are less persistent than HCFCs
- All Halogenated Hydrocarbons have very high global warming potential

Halogenated Hydrocarbons

Regulation

- CFCs
 - Phased out by the Montreal Protocol (in force 1 January 1989)
- HCFCs
 - Included in the Kyoto Protocol
 - Regulated under the Kigali Amendment to the Montreal Protocol

Halogenated Hydrocarbons

Regulation

- Halogenated Hydrocarbons are being replaced by hydrocarbon refrigerants (propane (R-290), iso butane (R-600A)) or CO₂ in some applications
 - For space cooling R-454B is being introduced which is a mixture of R-32, a hydrofluorocarbon (HFC), and R-1234yf a hydrofluoroolefin (HFO)
 - R-1234yf on its own is common in modern automotive air-conditioning systems

PFAS (per- and polyfluoroalkyl substances)

PFAS

Uses

- Surfactants, Lubricants, and Repellents for dirt, water, and grease
- Firefighting foams, Food packaging, Drugs, Cosmetics, Sunscreens, Pesticides, Textiles, Non-stick cookware, Vehicles, and Electronics

PFAS

Concerns

- These molecules are extremely non reactive and therefore persistent
- They are highly mobile in the environment moving through food and water cycles as well as be carried in dust and airborne particles
- Some molecules are easily absorbed
- Potential human health effects associated with PFAS include effects on the liver, kidney, thyroid, immune system, nervous system, metabolism and bodyweight, and reproduction and development
- Impacts on wildlife include toxicity to the immune and nervous systems, and general effects on growth, reproduction, and development

PFAS

Regulation

- Regulation of PFAS is emerging
- The Canadian regulations to address PFAS are taking a proactive approach to regulating pollutants
 - New substances must be reported before they are imported or manufactured at scale

Climate Change

Climate Change

The role of greenhouse gasses

- Greenhouse gasses is a label applied to any number of gasses that when present in the atmosphere reflect heat (infrared radiation) back towards the planet
- Greenhouse gasses are necessary for life as we know it
 - Without the greenhouse effect, nights would be extremely cold
- Human emissions have increased to concentration of greenhouse gasses in the atmosphere and therefore average global temperatures

Climate Change

Effects

- Increased temperatures may be seen as a warming in some areas
 - This is the origin of the term ‘Global Warming’
 - This is also being seen as extreme heat waves
- Increased frequency and severity of severe weather events
- Changes in rainfall patterns (droughts in some areas, flooding in others)
- Increased frequency and severity of wild fires
- Sea Level rise

Climate Change

Effects

- Changes in habitat suitability
 - This may result in population changes especially for plants and non-migratory species
- Longer frost free and growing seasons
- Reduced ice in arctic waters
 - Safer and more reliable polar navigation

Climate Change

Tipping points

- The term tipping point refers to events that are accelerated by climate change that result in the acceleration of climate change
- These include:
 - The release of CO₂ by climate change caused wildfires
 - The release of Methane from melting permafrost

Climate Change

Human Impacts

- Climate change can impact human populations in numerous ways:
 - Flooding of coastal area resulting loss of homes (climate migration)
 - Loss of food production due to flooding or drought
 - Increased food production due to longer growing seasons
 - Improved access to natural resources at arctic attitudes
 - Changes to forestry and forrest products production